

求积分

$$\int_0^1 \frac{\ln(1+t^\alpha)}{t} \mathrm{d}t$$

将被积函数展开得到

$$\begin{aligned} &\int_0^1 \frac{1}{t} \sum_{n=1}^\infty \frac{(-1)^{n+1}}{n} t^{\alpha n} \mathrm{d}t \\ &= \sum_{n=1}^\infty \frac{(-1)^{n+1}}{n} \int_0^1 t^{\alpha n} \mathrm{d}t \\ &= \frac{1}{\alpha} \sum_{n=1}^\infty \frac{(-1)^{n+1}}{n^2} \\ &= \frac{1}{\alpha} \left[\zeta(2) - \frac{1}{4} \zeta(2) \right] = \frac{1}{\alpha} \beta(2) \\ &= \frac{\pi^2}{12\alpha} \end{aligned}$$